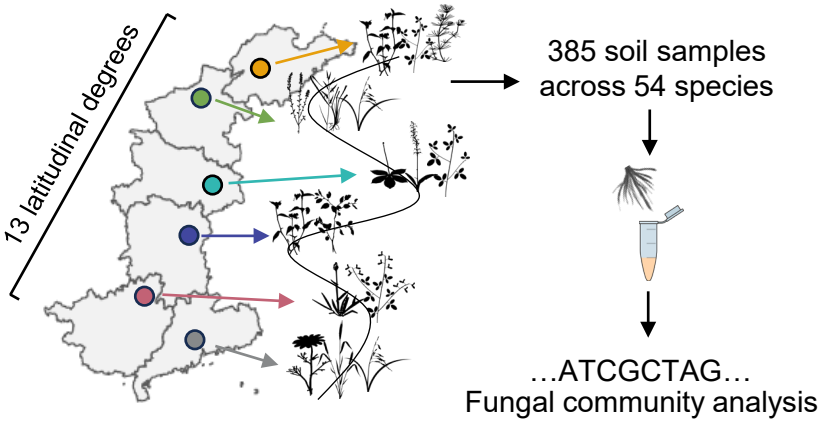
Step 1

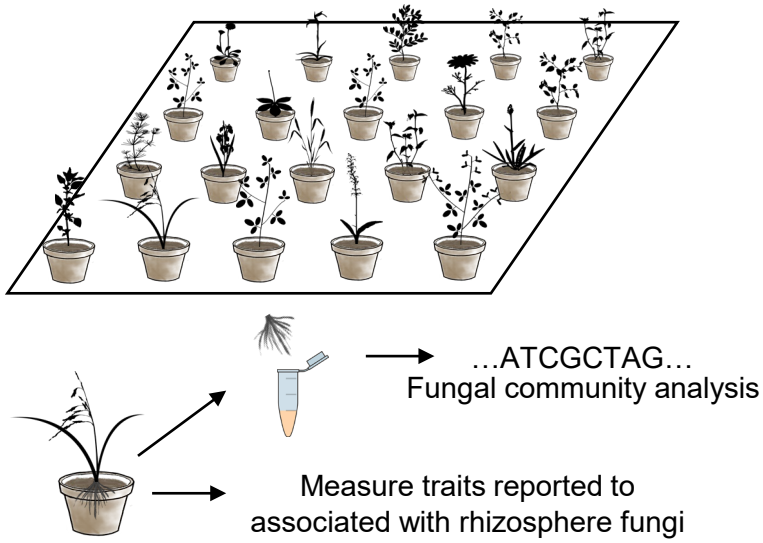
Field survey: variation in rhizosphere fungi among co-occuring plants in heterogeneous field conditions



Q1 : Do the composition of rhizosphere fungal communities of co-occurring plants respond differently to the same environmental changes?

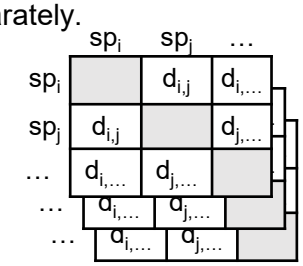
Step 2

Greenhouse expeirment: variation in rhizospehre fungi among species in a homogeneous condition



Step 3

For each species, we estimated its rhizosphere fungi compositional distinctiveness (Fungi-Dist) compared to that of all other species co-occurred at the same site and year, using the field (Fungi-Dist $_{\text{estimated in the field}}$) and greenhouse (Fungi-Dist $_{\text{estimated in greenhouse}}$) data separately.



$$\text{Fungi-Dist}_i = \frac{\sum_{j=1, i \neq j}^{N} d_{ij}}{N - 1}$$

N is the number of co-occurring plant species at a given site and year, d_{ij} is the community composition pair-wise distance between species *i* and *j*.

Step 4

For each species, we quantified the environmental effects on the compositional distinct of its rhizosphere fungi compared to that of all other species co-occurred at the same site and year.

$$\text{Environmental effects} = \text{Ln}\left(\frac{\text{Fungi-Dist}_{\text{estimated in the field}}}{\text{Fungi-Dist}_{\text{estimated in greenhouse}}}\right)$$

At species level: a positive and negative environmental effect means that the species recruits a more compositionally distinct or similar rhizosphere community in comparison to that of other species in the field versus in the greenhouse, respectively.

At community level (averaged across all co-occurring): a positive and negative mean environmental effect suggests that field environments increase and decrease compositional dissimilarity among co-occurring species rhizosphere communities, respectively.

Q2 : What are the directions and magnitudes of the environmental effects on the compositional variation in rhizosphere fungal communities among co-occurring plant species in nature?

Q3 : Which plant characteristics and environmental factors drive the environmental effects?